Shaheed Zulfikar Ali Bhutto Institute of Science & Technology

COMPUTER SCIENCE DEPARTMENT

Total Marks: 04

Obtained Marks: _____

Parallel and Distributed Computing

Assignment # 01

Submitted To: **Rana Faisal Hayat**

Student Name: **Ubaid Bin Waris**

Reg Number: **2212416**

Instructions: *Copied or shown assignments will be marked zero. Late submissions are not entertained in any case.*

Task1:

Write a 2-page analytical essay addressing:

- Why Amdahl's Law fails to model cloud or microservice systems accurately.
- How virtualization and containerization redefine "speedup" and "efficiency".
- Trade-offs between reliability, scalability, and power (three walls of HPC).

Task 2:

Scenario:

A research data center in 2025 runs large-scale simulations that perform 3.2×10^{17} FLOPs on a 64-node HPC cluster.

Each node delivers 1.5 TFLOP/s sustained and consumes 420 W on average during computation. Koomey's Law states that the number of computations per kilowatt-hour (kWh) doubles every 1.57 years.

Assume the baseline (2025) efficiency is 1.8×10^{14} FLOPs per kWh.

Compute the energy required in 2025

1. Compute the total sustained performance of the cluster (in FLOP/s).
2. Find the total execution time for 3.2×10^{17} FLOPs.
3. Calculate the total energy consumed (in kWh) and the cost if electricity is PKR 62 per kWh.

Apply Koomey's Law

1. Using Koomey's doubling rate, estimate the energy required for the same workload in 2028 and 2032.
2. Express the improvement in energy efficiency as a percentage compared to 2025.

Amdhal's law and Modern distributed system

Amdhal's law is a classical problem/concept in parallel computing and that explain the theoretical program speed up when multiple processors are used. It states the maximum improvement in performance is limited by the portion of the program when that must run sequentially. In simple terms, even if we add many processors the overall speedup cannot exceeds the limit by the non-parallel part of the program.

However, the Amdhal's law does not accurately model modern system such as cloud computing platforms and microservice architectures. One major reason that Amdhal's law assumes a fix workload and a single program with clearly define sequential and parall portions. Modern distributed systems are very different.

Another limitation is that Amdhal's law ignore overhead cause by distributed infrastructure. In cloud system, tasks may experience delays due to network latency, container scheduling or data transfer between services. These factors reduce the actual performance gain when more resources were added. Therefore amdhal's law becomes less useful for predicting the behaviour for distributed and elastics systems.

A more realistic approach in modern computing is gustafson's law which considers scable workloads instead of fixed work loads. Cloud system usually increase the size of the problem as more resources become available for example instead of finishing a job faster, organization may process

large data set or run more simulations at the same time.

Virtulization and Containerization Redefining speed up and Efficiency

Traditional performance matrix measure speed by comparing the runtime of the program on one processor versus multiple processors, However multiple processors, technologies such as virtulization and containerization have change how efficiency works.

Virtulization allows multiple virtual machine to run on a single physical server. Each virtual machine behaves like a an independent system with its own operating system. This improves resource ultilization because one server can support multipe workloads at the same time instead of focing only one execution speed, efficiency.

Containerization, which is widely used technologies like dockers and kubernetes. further improves efficiency now also includes how. containers share the whole operating system while isolating application for each other. Because containers are light weight compare to virtual machines.

In these environment, speed up is no longer measured only by faster execution of a single program. Instead, it includes higher system throughput meaning more tasks completed per unit of time. Efficiency also considers factors such as server utilization

development flexibility and the ability to scale services automatically. for example a containized system may not reduce the execution time of a single task, but it can process hundreds of tasks simultaneously using the same infrastructure.

Reliability, Scability and Power. The three wall of High Performance Computing

High Performance Computing (HPC) system face three major limitation often refered to as the three walls power wall, scability wall, reability wall.

The power wall refers to increasing amount of electrical power required by modern processors and large data centers. As computing performance increase, energy consumption also grows. cooling system and power delivery become major challenges.

The scability wall occurs when adding more processor no longer improves performance significantly communication overhead between node can increase, and some part of the application may still run sequentially.

Then reclibility wall becomes important in very large systems thousands or millions of component. The probability of hardware failures increase as the system size grows. Even small failure in one node can interrupt a large computation.

These three factors create trades-offs. Improving performance by adding more node may increase energy consumption. and failure probability. Designers of HPC system must carefully balance these factors to achive optimal performance.

Task # 2

Given Data

Total workload = 3.2×10^{17} FLOPs

No. of nodes = 64

Performace per node = 1.5 TFLOPs/s

1 TFLOPs = 10^{12} FLOPs

Power per node = 420 W

Electricity cost = 62 PKR per KWh

Baseline efficiency (2025) = 1.8×10^{14} FLOPs per KWh

Part A:

Energy Required in 2025

Total sustained performance of cluster = performance per node * no. of nodes

= 1.5 TFLOPs/s * 64

= 9.6×10^{13} FLOP's/s

Total execution time = Total FLOPs / cluster performance

$$= \frac{3.2 \times 10^{12}}{9.6 \times 10^{13}}$$

$$= 3333\ s \approx 0.925\ h$$

Total Energy consumption = total power of cluster * time

$$= 64 \times 420\ W \times 0.925$$

$$= 24.06\ kWh$$

Electricity cost = Total energy consumption * electricity cost

$$= 24.06 \times 62$$

$$= 1541.32\ PKR$$

Part B

Efficiency in 2020

time difference

2025 → 2020

3 Years

Doubling factor

$3/1.57 \approx 1.91$ doubling

Efficiency multiplier

$2^{1.91} \approx 3.76$

New efficiency

$1.0 \times 10^{14} \times 3.76$

6.77×10^{14} FLOPs / kWh

Energy required

$$\frac{3.2 \times 10^{17}}{6.77 \times 10^{14}} = 473 \text{ kWh}$$

Efficiency in 2032

2025 -> 2032 = 7 Years

Dobling factor

$7/1.57 = 4.46$

Efficiency multiplier

$2^{4.46} = 21.9$

New efficiency

$= 1.0 * 10^{14} * 21.9$

$= 3.94 * 10^{15}$ FLOPs / kWh

Energy required

$$\frac{3.2 * 10^{17}}{3.94 * 10^{15}} = 81 \text{ kWh}$$

Energy efficiency Improvement

2025 energy = 1770 kWh
2030 energy = 473 kWh
2032 energy = 81 kWh

Improvement by 2030
= 73% reduction in energy

Improvement by 2032
≈ 95% reduction in energy